

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fifth Semester B.E. (E&E) (Credit System) Degree Examinations

January 2023

20EE503 - LINEAR CONTROL SYSTEMS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Unit**.**Unit – I**

- | | Marks | BT* | CO* | PO* |
|---|--------------|------------|------------|------------|
| 1. a) Define control systems. Mention the differences between open loop and closed loop control systems. | 6 | L*1 | 1 | 1,2 |
| b) Define the following with respect to state space representation.
(i) State (ii) State variable (iii) State vector
(iv) Give the state space and output equation. | 4 | L1 | 1 | 1,2 |
| c) For the mechanical system shown in Fig. Q1(c),
(i) Write the nodal circuit (ii) Write the performance equation
(iii) Write force to voltage and force to current analogies circuits. | | | | |

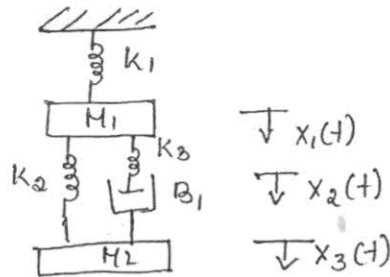


Fig. Q1(c)

- | | | | | |
|--|----|----|---|-----|
| 2. a) Define Transfer function. Also derive the transfer function relating displacement and excitation voltage for the armature controlled DC motor. | 10 | L2 | 1 | 1,2 |
| b) For the electrical network shown in Fig. Q2(b), obtain transfer function $\frac{V_o(s)}{V_i(s)}$ | 10 | L4 | 1 | 1,2 |

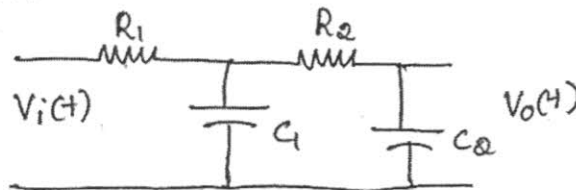


Fig. Q2(b)

- | | | | | |
|--|----|----|---|-----|
| 3. a) Define the following terms related to signal flow graph.
(i) Source node (ii) Sink node (iii) Forward path and its gain
(iv) Non touching loops (v) Feedback loop. | 10 | L1 | 2 | 1,2 |
| b) Derive the expressions for Peak Time (Tp) and Peak Overshoot in terms of ξ and ω_n for a second order control system. | 10 | L2 | 2 | 1,2 |
| 4. a) Draw the signal flow graph and obtain the Transfer function $\frac{X_5}{X_1}$ for the following set of linear equations.
$X_2 = a_{12}X_1 + a_{22}X_2 + a_{32}X_3$
$X_3 = a_{23}X_2 + a_{43}X_4$
$X_4 = a_{24}X_2 + a_{34}X_3 + a_{44}X_4$
$X_5 = a_{25}X_2 + a_{45}X_4$ | 10 | L3 | 2 | 1,2 |

P.T.O.

- b) Open loop Transfer function of a unity feedback system is given by $G(S) = \frac{K}{S(1+TS)}$, where K and T are positive constants. By what factor should the amplifier gain 'K' be reduced so that peak overshoot of a unit step response of the system is reduced from 75% to 25%.

10 L3 2 1,2

Unit – III

5. a) For the characteristic equation given below, determine the number of roots of the characteristic equation in the RHS of s-plane $S^6+3S^5+5S^4+9S^3+8S^2+6S+4=0$.
- b) A unity feedback control System is characterized by the open loop transfer function, $G(S) = \frac{K(S+13)}{S(S+3)(S+7)}$ using RH criteria (i) Calculate the range of K for the system to be stable (ii) Determine the value of K which will cause sustained frequency of oscillations in the closed loop System. What are the corresponding oscillation frequencies?
6. a) Define root locus. Explain the following. (i) Asymptotes (ii) Centroid (iii) Breakaway point with suitable example.
- b) Sketch the root locus of the control system with open loop transfer function $G(S)H(S) = \frac{K}{S^2+10S+100}$. Determine the stability of closed loop system.

10 L2 3 1,2

10 L3 3 1,2

10 L1 3 1,2

10 L2 3 1,2

Unit – IV

7. a) Define the following terms. (i) Bandwidth (ii) Gain cross over frequency (iii) Phase cross over frequency (iv) Phase margin (v) Gain margin.
- b) Find the open loop transfer function of the system whose approximate plot is shown Fig. 7(b).

10 L1 4 2,3

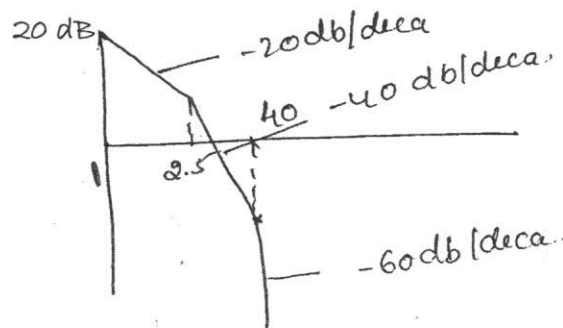


Fig. 7(b)

8. a) Explain step by step procedure to plot Bode plot.
- b) Explain Minimum and Non minimum phase system.

10 L3 4 2,3

10 L1 4 2,3

10 L1 4 2,3

Unit – V

9. a) The open loop transfer function of a single loop system is given by $G(s) = \frac{K}{S(S+1)^2}$. Obtain the Nyquist plot and comment on stability of system.
- b) Explain the design procedure for PID controller with suitable diagram.
10. a) With neat circuit, explain lead lag compensation network.
- b) Construct the Nyquist plot for the control system with $G(S)H(S) = \frac{K}{S(S+2)}$. Find the range of K for which control System is stable.

10 L4 5 4

10 L1 5 2,3

10 L1 5 2,3

10 L2 5 2,3